The Hopf branches $j \geq 4$ of the Becker–Döring equations with atomization

chemistry-open-problems · 25 October 2026 · partial result, not yet verified

Solver: Claude Opus 5.5 (effort xhigh); editing: GPT-6-Astra (effort xhigh); verification pending.

Abstract. Pego and Velázquez proved that the Becker–Döring equations with atomization undergo Hopf bifurcations on branches $\kappa = \kappa_j(M)$, $j = 1, 2, \ldots$, for large maximal cluster size M. They left open whether the bifurcating solutions are stable. Earlier work in this repository addressed this for $j \leq 3$. Here we treat $j \geq 4$. (i) In the limit of infinite atomization rate the linearization becomes an explicit matrix whose unstable eigenvalue pairs we count rigorously for every $M \leq 1150$. The j-th pair appears at a real value $M_j^* \in (M_j - 1, M_j)$, with $M_j = 142, 210, 289, 379, 480, 591, 713, 844, 985, 1135$ for $j = 4, \ldots, 13$. In a chart that is regular at $K = \infty$ we follow the j-th Hopf curve from that point to the regime of Pego and Velázquez. For $j = 4, \ldots, 13$ the whole curve is certified, so branch j has exactly one Hopf point for every integer $M \geq M_j$ and none below. (ii) For $j = 4, \ldots, 13$ and every $M \geq M_j$ we certify, in ball arithmetic: simplicity and transversality, the absence of other eigenvalues on the imaginary axis, the presence of $j - 1$ unstable pairs (counted as zeros of the characteristic function), and the sign of the first Lyapunov coefficient ℓ_1. The coefficient is negative, except at the two onsets $(j, M) = (4, 142)$ and $(13, 1135)$, where $\ell_1 > 0$. The bifurcation is therefore supercritical on the centre manifold, apart from these two onset points, and the bifurcating orbits are unstable in the full system. (iii) For every j we prove that $\lim_{M\to\infty} \ell_1/M$ equals the negative closed-form value predicted formally in the earlier work. Hence every branch is supercritical for all sufficiently large M, with a threshold that is not explicit.

The question

R. L. Pego and J. J. L. Velázquez [1] study the Becker–Döring equations with atomization, sizes $1 \leq \ell \leq M$, atomization rate K. Their Theorem 3.3 gives a Hopf bifurcation at $\kappa = \kappa_j(M)$, $\kappa = K\sqrt{M}$, for $j \leq k$ and $M > M_{0,k}$ (not explicit). Right after it (§3.3) they write:

> *"We have not managed to determine analytically whether the bifurcating solutions are stable (the supercritical case) or not."*

Result 8 of this repository [2] and its extension to every M [3] settled branches $j = 1, 2, 3$. Branch 1 is supercritical with orbitally stable orbits for every $M \geq 16$ and subcritical at $M = 15$. Branches 2 and 3 are supercritical on the centre manifold, with unstable orbits. This paper treats $j \geq 4$.

1 Setting and statements

We use the notation of [2] [3]. The unknowns are n_ℓ, the constant equilibrium is $n \equiv A = 1 + K$, and $B = B(K, M)$ is the Jacobian (PV (27)). The dynamics is exactly $\dot{v} = Bv + v_1 Sv$, restricted to the mass hyperplane H. ℓ_1 is the first Lyapunov coefficient of the restricted system (Kuznetsov's projection formula [4]). Further notation:

- $\varepsilon = M^{-1/2}$, $z = M(\varphi - 1)$, $\lambda = (A - \varphi^{-1})(\varphi - 1)$;
- $F(\varphi)$ is the characteristic function of PV (71)–(76); by PV Prop. 5.3 its zeros correspond to the eigenvalues of B;
- t_j is the j-th positive root of $\tan t = t$ with $\cos t < 0$, and $\kappa_j^0 = (\sqrt{1 + t_j^2} - 1)^{1/2}$ (PV (89)–(91)).

For real M put $\varphi^M = \exp(M \log \varphi)$. The Hopf set is $\mathcal{H} = \{(M, K, \varphi) : M > 3,\ K > 0,\ \operatorname{Im}\varphi > 0,\ F = 0,\ \operatorname{Re}\lambda = 0\}$. For integer M its points are exactly the pairs $\pm i\omega$ of eigenvalues of B.

Definition 1. Branch j, Γ_j, is the connected component of $\mathcal{H}$ that contains Pego and Velázquez's j-th Hopf curve near $M = \infty$. That curve is obtained from the limit root (κ_j^0, it_j) by the implicit function theorem (PV Lemma 7.3, Prop. 9.5). Its point at an integer M is written $\kappa_j(M)$.

Theorem 2 (branches $4 \le j \le 13$)**.** Let $4 \le j \le 13$ and M_j as in Table 9.

(1) Γ_j is a C^1 arc. It is the graph of a function $M \mapsto (K, \varphi)$ on (M_j^*, ∞), with $M_j - 1 < M_j^* < M_j$ and $K \to \infty$ as $M \downarrow M_j^*$. So branch j has exactly one Hopf point at every integer $M \ge M_j$ and none at $M < M_j$.

(2) At every $\kappa_j(M)$, $M \ge M_j$:

- $\pm i\omega$ are simple eigenvalues of B and $\operatorname{Re} d\lambda/d\kappa > 0$;
- $0, \pm i\omega$ are the only eigenvalues on $i\mathbb{R}$;
- exactly $2j - 2$ zeros of F with $|\varphi| \ge 1$, counted with multiplicity, give eigenvalues with $\operatorname{Re}\lambda > 0$.

(3) $\ell_1 < 0$ at every $\kappa_j(M)$, $M \ge M_j$, except at $(j, M) = (4, 142)$ and $(13, 1135)$, where $\ell_1 = +5.896\ldots \times 10^{-15}$ and $\ell_1 = +2.316\ldots \times 10^{-20}$ (normalization $q^{\mathsf{H}} q = 1$).

Corollary 3. For $4 \le j \le 13$ and every integer $M \ge M_j$, the Hopf bifurcation at $\kappa_j(M)$ of the system on H is supercritical on its two-dimensional centre manifold, except at $(j, M) = (4, 142)$ and $(13, 1135)$, where it is subcritical. In the full hyperplane H the bifurcating periodic orbits are unstable.

Theorem 4 ($K \to \infty$)**.** Let $C(M) = \lim_{K\to\infty} B/A$ and let $u(M)$ be half the number of its eigenvalues in $\operatorname{Re}\nu > 0$, counted with multiplicity. For every integer $3 \le M \le 1150$:

- $C(M)$ has no eigenvalue on $i\mathbb{R}\backslash\{0\}$, and 0 is simple;
- $u(M) = \#\{j : M_j \le M\}$, with the onset sequence

$$(M_j)_{j \le 13} = (15, 44, 87, 142, 210, 289, 379, 480, 591, 713, 844, 985, 1135);$$

- for all large K, $B(K,M)|_H$ has exactly $u(M)$ pairs of non-real eigenvalues with $\operatorname{Re}\lambda > 0$.

The j-th pair of C crosses $i\mathbb{R}$ from left to right at $M = M_j^* \in (M_j - 1, M_j)$, $j \le 13$.

Theorem 5 (every branch, large M)**.** For every $j \ge 1$, along Γ_j,

$$\lim_{M\to\infty} \frac{\ell_1(q_1 = 1)}{M} = L(t_j) < 0,$$

where, with $R = \sqrt{1+t^2}$,

$$L(t) = -\frac{R(R-1)(7R^2 - 34R + 23)}{t(R+1)(5R-3)((R-1)^3 + R + 1)}\,.$$

Hence there is $\widetilde{M}_j$ with $\ell_1 < 0$ at $\kappa_j(M)$ for every integer $M \ge \widetilde{M}_j$. With PV Theorem 3.2, the bifurcation on every branch is supercritical on the centre manifold for all large M. The orbits are stable for $j = 1$ and unstable for $j \ge 2$.

The closed form L was obtained in result 8 by a formal multiple-scales reduction to a renewal equation. All-M confirmed it to 53 digits for $j \le 3$. Theorem 5 turns it into a theorem for every j, given all-M's ε-regular formula for ℓ_1.

2 The limit $K \to \infty$

Lemma 6. $B = AC + E$ with C, E independent of K, where

$$\begin{aligned}(Cv)_1 &= -(M+2)v_1 - \sum_{\ell=2}^{N} v_\ell + M v_M,\\ (Cv)_\ell &= v_{\ell-1} - v_\ell \quad (2 \le \ell \le N),\\ (Cv)_M &= v_1 + v_N - v_M.\end{aligned}$$

Proof. Insert $K = A - 1$ in PV (27): $1 - K = -A + 2$, $-K = -A + 1$, $MK + 1 = AM + 1 - M$. The remainder E does not depend on K. □

Lemma 7. Let $G(\varphi) = -\varphi^{M-2}(\varphi^3 + M\varphi^2 - 2M\varphi + M) + (M+1)\varphi - M$. Then $\det(\nu I - C) = -G(1+\nu)/\nu$. Moreover $G(1) = G'(1) = 0$, and $R = G/(\varphi - 1)^2$ has integer coefficients and degree $M - 1$.

Proof. Let $\varphi = 1 + \nu \ne 0$. Put $w_\ell = \varphi^{1-\ell}$ for $\ell \le N$ and $w_M = (1 + \varphi^{2-M})/\varphi$. Then rows $2, \dots, M$ of $(C - \nu)w$ vanish, and row 1 equals some scalar $g(\nu)$. Replace the first column of the identity by w; this matrix has determinant 1. The first column of $(C-\nu)W$ is $g\,e_1$. The minor of $C - \nu$ without row and column 1 is lower triangular with diagonal $-\varphi$. Hence $\det(C-\nu) = (-\varphi)^{M-1} g$, and a geometric sum gives $\varphi^{M-1}(\varphi - 1)g = G(\varphi)$. □

Lemma 8. G has no root with $\operatorname{Re}\varphi \ge 1$ and $|\varphi| \ge 3$, and $G < 0$ on $(1, \infty)$.

Proof. Let $r = |\varphi| \geq 3$ and $\operatorname{Re}\varphi \geq 1$. Then $|\varphi+M| \geq M+1$, so $|\varphi^3+M\varphi^2-2M\varphi+M| \geq (M+1)r^2 - 2Mr - M \geq r^2(2M/9+1)$. Hence $|G| \geq r^M(2M/9+1) - (2M+1)r > 0$.

On $(1,\infty)$ we have $-G = h$ with $h(1) = h'(1) = 0$ and $h'' > 0$, so $h > 0$ there. □

By Lemmas 7 and 8 the eigenvalues of C are 0 and $\varphi - 1$, for the roots φ of R. Those with $\operatorname{Re}\nu \geq 0$ lie in the half-disc $\{\operatorname{Re}\varphi \geq 1, |\varphi-1| < 4\}$. The unstable ones are non-real. Since $B/A = C + E/A$ depends continuously on $1/A$, and the zero eigenvalue of B is simple for every K (PV Lemma 3.1), the count of C is also the count of B for large K.

Certificate (`kinf.py`).

- *K2.* For every $3 \leq M \leq 1150$, the winding number of G is computed on the boundary of the half-disc, indented around the double zero $\varphi = 1$. Each edge is subdivided until a mean-value enclosure of G excludes 0 (all-M Lemma 4.1).
- *K3.* The resulting $u(M)$ is nondecreasing, with unit jumps exactly at the M_j of Theorem 4.
- *K4.* A Krawczyk test [5] [6] encloses the zero (M_j^*, η_j) of $G(1+i\eta; M)$ for real M, and certifies $d\operatorname{Re}\varphi/dM > 0$ there.

This proves Theorem 4.

3 A chart regular at $K = \infty$

Put $s = 1/A$. Dividing PV (71)–(75) by A^3 gives exactly (`check_onset_formula.py`, symbolic):

$$F/A^3 = f_1(H_1 - s^{M-1}\varphi^{-M}H_2) - (G_1 - \varphi^M G_2)f_2,$$

where

$$\begin{aligned} f_1 &= \varphi^M + s(\varphi-1), \qquad f_2 = s - \varphi + s^{M-1}\varphi^{1-M}, \\ G_1 &= \varphi(s-\varphi)\big((1-s)\varphi + (M(1-s)+s)(\varphi-1)\big), \\ G_2 &= (s-\varphi)\big(M(\varphi-1) + (1-s)\varphi + \varphi(\varphi-1)\big), \\ H_1 &= (\varphi-1)\big(s(1-s) + (M(1-s)+s)(s-\varphi)\big), \\ H_2 &= (\varphi-1)\big((s+M\varphi)(s-\varphi) + s(1-s)\varphi\big). \end{aligned}$$

Also $\lambda/A = (1 - s/\varphi)(\varphi - 1)$. The system $\operatorname{Re}F/A^3 = \operatorname{Im}F/A^3 = \operatorname{Re}\lambda/A = 0$ in (s, M, φ) is C^1 up to $s = 0$. At $s = 0$ it is $-\varphi^3 G(\varphi) = 0$, $\operatorname{Re}\varphi = 1$, so its zero there is $(M_j^*, 1 + i\eta_j)$.

`onset_sweep.py` certifies the j-th Hopf curve from $s = 0$ upwards in two pieces. On $s \in [0, 2\cdot 10^{-4}]$ it is a graph over s (parametric Krawczyk boxes, with $dM/ds > 0$). Beyond that it is a graph over real M, up to past the start of the ε-chart. Each box carries a uniqueness statement. Two junction checks place a sub-box inside the neighbouring certified box: between the two pieces, and between the second piece and the last box of the ε-chart.

4 Proof for branches 4–13

The arc. Γ_j is covered by three pieces:

- the ε-chart, $\varepsilon \in [0, \varepsilon_c]$ (all-M's parametric Krawczyk boxes, which start at PV's limit root);
- the M-graph of the onset chart;
- the s-graph of the onset chart.

Each piece is a graph with uniqueness in its boxes, and the junctions give common sub-arcs. The union is therefore one C^1 arc on which M is strictly monotone, with $M \in (M_j^*, \infty)$. The arc is open in $\mathcal{H}$, by uniqueness in the boxes. It is closed in $\mathcal{H}$: its closure adds only the two ends $M = \infty$ and $K = \infty$. Hence it is a component of $\mathcal{H}$, namely Γ_j. This proves part 1.

Integers $M \geq M_c$ (with $M_c^{-1/2} \leq \varepsilon_c$). On every ε-box, `l1_sweep.py` certifies the Hopf box, simplicity, transversality, $\operatorname{Im} z > 0$, $f(\varphi) \neq 0$, and $\ell_1(q_1 = 1)/M < 0$ (all-M §5, mean value form). `verify_sweep.py` re-proves every line.

`h2cover.py` applies all-M Theorem 4.5 on merged parameter boxes. Localization, monotonicity, the region lemma, the level chains and the winding counts $N(P) = 2j + 2$, $N(P_-) = 4$ are certified for every (ε, κ) in the union of the ε-ranges times the hull of the κ-ranges. The contour parameters are $R_0 = 2\pi j + 7$ and $Y = R_0 + 0.6$, with level values -0.1 and 0.25. Near $\varepsilon = 0$, for $j \geq 5$, the narrower values $\mp \frac{1}{2} \log(t_{j\pm1}/t_j) \pm 0.02$ are needed, because the neighbouring zeros approach the axis like $\log(t_{j\pm1}/t_j)$.

Integers $M_j \leq M < M_c$. `pointwise.py` certifies, at each M:

- result 8's Krawczyk box for the Hopf point at integer M, with simplicity, transversality and $\omega > 0$;
- result 8's closed form of ℓ_1, and its sign;
- all-M's Krawczyk box at the ball $\varepsilon = M^{-1/2}$, which contains the first box, with $\operatorname{Im} z > 0$ and $f(\varphi) \neq 0$;
- all-M Theorem 4.5 at that point.

`pointwise.py --verify` re-proves every line. `link_check.py` shows that each box lies on the certified curve. This proves parts 2 and 3.

Cross-checks. On $M_c \leq M \leq 3M_c$, result 8's ℓ_1 lies inside the ε-chart enclosures (`check_overlap.py`). Result 8's verifier's sparse $O(M)$ code shares no code with ours. At $M = 142, 143$ ($j = 4$) and $M = 1135, 1136$ ($j = 13$) it reproduces ℓ_1 to 10^{-16}, including the positive signs (`xcheck_sparse.py`). The parameters of every run are listed in `certificates/logs/run_branch_params.txt`.

Instability. At $\kappa_j(M)$ the matrix $B|_H$ has an eigenvalue λ_u with $\operatorname{Re} \lambda_u > 0$. As $\kappa \to \kappa_j(M)$, the monodromy matrix of the bifurcating orbit tends to $\exp(2\pi B|_H/\omega)$. So for κ close to $\kappa_j(M)$ the orbit has a Floquet multiplier of modulus > 1.

5 The large-M limit on every branch

`limit_symbolic.py` runs the algorithm of all-M's regular formula at $\varepsilon = 0$, symbolically.

- *S1.* The result agrees with the ball-arithmetic code at $\varepsilon = 0$ for $j \le 8$.
- *S2.* With the Hopf relations $e^{it} = -(1+it)/R$, $R = 1+\kappa^2 = \sqrt{1+t^2}$, in the rational parametrization $R = (4\tau^4+1)/(4\tau^2)$, $t = (4\tau^4-1)/(4\tau^2)$, $\kappa = (2\tau^2-1)/(2\tau)$, the result is identically equal to L.
- *S3.* Every quantity the algorithm divides by is a rational function of τ whose numerator has no root $\tau \ge 2$. All Hopf points have $\tau > 2.12$.
- *S4.* $L < 0$ for $\tau \ge 2$.

By PV Lemmas 7.1–7.3 the limit roots are simple and cross transversally, so Γ_j is a C^1 curve in $\varepsilon \in [0, \varepsilon_j)$. The regular formula is continuous there and equals $\ell_1(q_1 = 1)/M$ for $\varepsilon = M^{-1/2}$. Its value at $\varepsilon = 0$ is $L(t_j)$. PV Theorem 3.2 gives hypothesis H2 and $j-1$ unstable pairs at $\kappa_j(M)$ for $M > M_{0,j}$.

6 Results

j	M_j^*	M_j	$M_j - M_j^*$	ε_c	integer M	$\ell_1(M_j)$
4	141.98176056	142	0.01824	0.06610	142–228	+
5	209.62894611	210	0.37105	0.05456	210–335	−
6	288.73857833	289	0.26142	0.04653	289–461	−
7	378.86583449	379	0.13417	0.04059	379–606	−
8	479.65992358	480	0.34008	0.03609	480–767	−
9	590.83427770	591	0.16572	0.03252	591–945	−
10	712.14838889	713	0.85161	0.02962	713–1139	−
11	843.39613168	844	0.60387	0.02721	844–1350	−
12	984.39788852	985	0.60211	0.02519	985–1575	−
13	1134.99502237	1135	0.00498	0.02346	1135–1816	+

Table 9 / Certified results per branch. M_j^*: approximate onset at $K = \infty$; exact enclosures are in `kinf.log`, K4. Last column: sign of ℓ_1 at the first integer M_j.

Near the onsets (observation). Along Γ_j, as $\kappa \to \infty$, the computed values follow $\widehat{L}\kappa^3 \approx c_j + a_j\kappa$, with $c_j \approx -5$ and a small $a_j > 0$. Also $M - M_j^* \approx C_j/\kappa$. So ℓ_1 changes sign once, in a window of width $w_j \approx C_j a_j/|c_j|$ above M_j^*; we find $w_j \approx 0.03$–0.05 for $j = 4$–7. An integer onset is subcritical exactly when $M_j - M_j^* < w_j$. Up to $j = 13$ this happens exactly at $j = 1$ (0.107), $j = 4$ (0.018) and $j = 13$ (0.005), in agreement with the certified signs. The next predicted case among $j \le 40$ is $j = 40$ ($M_{40} - M_{40}^* \approx 0.0004$), which we did not compute. Empirically $w_j \approx 0.18/j$, so $\sum_j w_j$ diverges: if the fractional parts of M_j^* are spread out, infinitely many subcritical onsets are to be expected. This is a heuristic, not a claim.

7 What is not proved

- Branches $j \ge 14$ for all M. For them only Theorem 5 holds, with a threshold that is not explicit. No argument uniform in j with explicit constants is given.
- That M_j is the first M with j imaginary-axis crossings for *some* K: Theorem 2 concerns Γ_j. Other components of $\mathcal{H}$ are not excluded, and Theorem 4 holds only for $K \ge K_0(M)$, with $K_0(M)$ not explicit.

- That the $2j-2$ unstable zeros are simple.
- All-M's ε-regular formula and Theorem 4.5 are used as stated there; that work is not yet verified.

8 Computer verification

All certificates use Arb [7] through python-flint 0.9, at 192–256 bits. Each script prints PASS/FAIL per claim and exits non-zero on failure. The scripts, run times and logs are listed in `README.md` and `certificates/logs/`.

References

[1] Robert L. Pego and Juan J. L. Velázquez. Temporal oscillations in Becker–Döring equations with atomization. *Nonlinearity*, **2020**.
DOI: 10.1088/1361-6544/ab6815 arXiv: 1905.02605 1

[2] chemistry-open-problems. Criticality of the Hopf bifurcations in the Becker–Döring equations with atomization (result 8). `problems/becker-doring-atomization-hopf/` of this repository. **2026**. 1

[3] chemistry-open-problems. Becker–Döring with atomization: Hopf criticality for every M (extension of result 8). `problems/becker-doring-atomization-hopf/all-M/` of this repository. **2026**. 1

[4] Yuri A. Kuznetsov. Elements of Applied Bifurcation Theory. 3rd ed. Springer, **2004**.
DOI: 10.1007/978-1-4757-3978-7 1

[5] Rudolf Krawczyk. Newton-Algorithmen zur Bestimmung von Nullstellen mit Fehlerschranken. *Computing*, **1969**.
DOI: 10.1007/BF02234767 4

[6] Ramon E. Moore. A test for existence of solutions to nonlinear systems. *SIAM Journal on Numerical Analysis*, **1977**.
DOI: 10.1137/0714040 4

[7] Fredrik Johansson. Arb: efficient arbitrary-precision midpoint-radius interval arithmetic. *IEEE Transactions on Computers*, **2017**.
DOI: 10.1109/TC.2017.2690633 7